

Performance Testing

Date	08 July 2026
Project	Rising Waters – Flood Prediction System
Maximum Marks	5 Marks

Step 1: Testing Overview

Field	Details
Testing Tool	Flask Test Client, Postman
Type	Load Testing, Functional Performance
Target Module	Flood Prediction API
Environment	Local Flask Server (Python 3.11)
Test Date	08 July 2026

Step 2: Test Scenarios

Scenario	Users	Duration	Expected
Single prediction	10	30 sec	Pass
Concurrent requests	50	60 sec	Stable
Large dataset	20	45 sec	Processed
Invalid input	10	20 sec	Validation shown

Step 3: Performance Results

Metric	Target	Actual	Status
Avg Response	<2 s	1.2 s	Pass
Max Response	<5 s	2.8 s	Pass
Throughput	>20 req/s	28 req/s	Pass
Error Rate	<1%	0.2%	Pass
CPU	<80%	58%	Pass
Memory	<80%	61%	Pass

Observations

Key Findings: The application remained stable under normal and moderate load. Flood predictions were generated quickly with low error rates.

Bottlenecks: Slight increase in response time during simultaneous requests above 50 users.

Optimization: Optimized preprocessing, cached model loading, reduced redundant computations.